

Paper 2 - Set 13 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	C	Quadratic functions
2	A	Integration
3	A	Indices and surds
4	A	Identifying errors in proofs
5	E	Mathematical proof
6	A	Identifying errors in proofs
7	E	The logic of arguments
8	C	The logic of arguments
9	A	Mathematical proof
10	A	Mathematical proof
11	D	Pythagoras, mensuration and 3D
12	H	Sequences and series
13	E	Chains of reasoning
14	H	Number, units and estimation
15	B	Exponentials and logarithms
16	A	Identifying errors in proofs
17	H	Number, units and estimation
18	D	Chains of reasoning
19	H	Quadratic functions
20	H	Binomial expansion

Worked solutions

Question 1 QUA-15 · Quadratic functions · spec MM1.3

The graph of $y = x^2 + bx + c$ crosses the x -axis at $(1, 0)$ and $(5, 0)$. What is the y -coordinate of the vertex?

Answer: C

$$y = (x - 1)(x - 5), \text{ vertex at } x = 3: y = (2)(-2) = -4.$$

Question 2 INT-10 · Integration · spec MM7.2

Find $\int \left(x + \frac{1}{x}\right)^2 dx$.

Answer: A

$\left(x + \frac{1}{x}\right)^2 = x^2 + 2 + x^{-2}$, which integrates to $\frac{x^3}{3} + 2x - x^{-1} + c$. (Option E is a common error: you cannot integrate a power of a bracket that way.)

Question 3 IND-03 · Indices and surds · spec MM1.1

Solve $4^{x+1} = 8^{2x-3}$.

Answer: A

Write both sides as powers of 2: $2^{2x+2} = 2^{6x-9}$. So $2x + 2 = 6x - 9$, giving $4x = 11$ and $x = \frac{11}{4}$.

Question 4 ERR-16 · Identifying errors in proofs · spec Err2

A student argues: 'Every prime number is odd, because if it were even it would be divisible by 2 and so not prime.' Which of the following is correct?

Answer: A

A number divisible by 2 is not prime unless it equals 2. The conclusion 'every prime is odd' is false.

Question 5 PRF-01 · Mathematical proof · spec Prf1

Consider the claim: '**for every positive integer n , $n^2 + n$ is even.**' Which of the following is a complete and valid proof?

Answer: E

A covers every n with a general argument. B checks only three cases; C handles only even n ; D is false for odd n ; E is incomplete.

Question 6 ERR-02 · Identifying errors in proofs · spec Err2

Here is a purported proof that $1 = 2$.

- I Let $a = b$, with $a \neq 0$.
- II Then $a^2 = ab$.
- III So $a^2 - b^2 = ab - b^2$, i.e. $(a - b)(a + b) = b(a - b)$.
- IV Dividing both sides by $(a - b)$: $a + b = b$.
- V So $2b = b$, and hence $2 = 1$.

On which line does the first error occur?

Answer: A

Since $a = b$, $a - b = 0$ and dividing by zero is invalid.

Question 7 LGC-24 · The logic of arguments · spec Arg4

Let S be the statement: **every even number greater than 2 can be written as the sum of two primes.** A single counterexample to S would be

Answer: E

To refute 'for all n , $P(n)$ ' you exhibit one n with $P(n)$ false.

Question 8 LGC-08 · The logic of arguments · spec Arg4

Which of the following is the negation of **'there exists a positive integer n such that $n^2 + n + 41$ is not prime'**?

Answer: C

Negating 'there exists n with not $P(n)$ ' gives 'for all n , $P(n)$ '.

Question 9 PRF-16 · Mathematical proof · spec Prf4

A proof that **if a and b are positive real numbers then $\frac{a+b}{2} \geq \sqrt{ab}$ ** is given with its lines scrambled.

- I Hence $a + b \geq 2\sqrt{ab}$, and dividing by 2 gives the result.
- II Expanding, $a - 2\sqrt{ab} + b \geq 0$.
- III Since squares are non-negative, $(\sqrt{a} - \sqrt{b})^2 \geq 0$.

Which order gives a valid proof?

Answer: A

Start from a known truth (III), expand (II), rearrange (I). Starting from the result and working backwards (I, II, III) would need every step to be reversible and is presented in the wrong direction.

Question 10 PRF-03 · Mathematical proof · spec Prf1

Here is a proof that $\sqrt{2}$ is irrational, with its steps in the wrong order.

- I Then $p^2 = 2q^2$, so p^2 is even, so p is even; write $p = 2k$.
- II Suppose, for contradiction, that $\sqrt{2} = \frac{p}{q}$ where p and q are positive integers with no common factor.
- III This contradicts the assumption that p and q have no common factor, so $\sqrt{2}$ is irrational.
- IV Then $4k^2 = 2q^2$, so $q^2 = 2k^2$ is even, so q is even.

Which is the correct order?

Answer: A

Assume (II), deduce p even (I), deduce q even (IV), reach the contradiction (III).

Question 11 X2-29 · Pythagoras, mensuration and 3D · spec M5.16

P : 'a solid cylinder and a solid cube have equal volumes'. Q : 'the cylinder and the cube have equal surface areas'. Which one of the following is true?

Answer: D

Equal volumes say nothing about surface areas (a long thin cylinder has a large surface area), and equal surface areas say nothing about volumes. Neither implies the other.

Question 12 X2-23 · Sequences and series · spec MM2.2

An arithmetic sequence has first term a and common difference d , both non-zero. Which of the following statements is/are true?

- I If $d > 0$ then every term after some point is positive.
- II The sum of the first n terms is a quadratic function of n .
- III If a and d are integers then S_n is an integer for every n .

Answer: H

I: terms $a + (n - 1)d$ eventually exceed any bound. II: $S_n = \frac{n}{2}(2a + (n - 1)d)$ is quadratic in n . III: S_n is a sum of integers.

Question 13 RSN-19 · Chains of reasoning · spec Prf5

A bag contains red and blue balls only. If one red ball is removed, one quarter of the remaining balls are red. If instead two blue balls are removed, one third of the remaining balls are red. How many balls are in the bag?

Answer: E

With r red and b blue: $r - 1 = \frac{1}{4}(r + b - 1)$ and $r = \frac{1}{3}(r + b - 2)$. The second gives $b = 2r + 2$; substituting into the first, $4r - 4 = 3r + 1$, so $r = 5$, $b = 12$: 17 balls.

Question 14 NUM-12 · Number, units and estimation · spec M2.3

n is a positive integer such that n , $n + 2$ and $n + 4$ are all prime. Which of the following statements is/are true?

- I There is exactly one such value of n .
- II n must be odd.
- III One of n , $n + 2$, $n + 4$ must be divisible by 3.

Answer: H

Among any three numbers of the form n , $n + 2$, $n + 4$ one is a multiple of 3 (they cover all residues mod 3) - III. So one of them must equal 3 itself for all to be prime, giving $n = 3$ only (3, 5, 7) - I. And 3 is odd - II.

Question 15 X2-22 · Exponentials and logarithms · spec MM5.3

Consider the statement: **for all real x , $2^x > x^2$.** Which of the following is a counterexample?

Answer: B

$2^3 = 8 < 9 = 3^2$, so $x = 3$ breaks the claim. The other options satisfy it: $2 > 1$, $32 > 25$, $1 > 0$, $1024 > 100$.

Question 16 ERR-15 · Identifying errors in proofs · spec Err1

Claim: the sum of the first n odd numbers is n^2 .

- I The first odd number is $1 = 1^2$, so the claim holds for $n = 1$.
- II Suppose the claim holds for some n , i.e. $1 + 3 + \dots + (2n - 1) = n^2$.
- III Adding the next odd number, $2n + 1$, gives $n^2 + 2n + 1 = (n + 1)^2$.
- IV So the claim holds for $n + 1$, and hence for all positive integers n .

Which of the following is correct?

Answer: A

This is a valid proof by induction: a base case, and a step showing that truth for n implies truth for $n + 1$.

Question 17 X2-18 · Number, units and estimation · spec M2.3

n is a positive integer greater than 1. Which of the following statements is/are true?

- I If n is not prime, then n has a prime factor at most $\sqrt{n}$.
- II If n has no prime factor at most $\sqrt{n}$, then n is prime.
- III If n is prime, then n has no factor between 1 and n exclusive.

Answer: H

If $n = ab$ with $1 < a \leq b$ then $a \leq \sqrt{n}$ and a has a prime factor $\leq \sqrt{n}$ (I). II is the contrapositive of I. III is the definition of prime.

Question 18 RSN-03 · Chains of reasoning · spec Prf5

Five friends each shake hands with some of the others (each pair shakes hands at most once). Which of the following is impossible?

Answer: D

If four people each shook hands with all four others, everyone shook hands with everyone, so the fifth person also made 4 handshakes, not 1. The others can happen: B is a pentagon of handshakes, C is everyone with everyone, D is a chain, E leaves one person out.

Question 19 QUA-06 · Quadratic functions · spec MM1.3

The function $f(x) = x^2 - 6x + 13$ is defined for all real x . Which of the following statements is/are true?

- I $f(x) > 0$ for all real x
- II The graph of $y = f(x)$ is symmetric about the line $x = 3$
- III The equation $f(x) = 4$ has exactly one real solution

Answer: H

$f(x) = (x - 3)^2 + 4 \geq 4 > 0$, so I is true. The axis of symmetry is $x = 3$, so II is true. $f(x) = 4$ only when $(x - 3)^2 = 0$, i.e. $x = 3$: exactly one solution, so III is true.

Question 20 X2-13 · Binomial expansion · spec MM2.4

n is a positive integer. Which of the following statements is/are true?

- I The coefficient of x^n in $(1 + x)^{2n}$ is $\binom{2n}{n}$.
- II The sum of the coefficients in the expansion of $(1 - x)^n$ is 0.
- III $(1 + x)^n \geq 1 + nx$ for all $x \geq 0$.

Answer: H

I is the general term with $r = n$. II: put $x = 1$. III: expanding $(1 + x)^n$ gives $1 + nx$ plus non-negative terms when $x \geq 0$.